# When Does Non-Uniform KV Cache Quantization Help? Cardinality Constraints and a Heterogeneity Diagnostic

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## Abstract

Across seven models spanning 1.4B–14B parameters and five architecture families (MHA through GQA 8:1), we find that per-head non-uniform KV cache quantization on hardware-aligned bit-widths $\{2, 4, 8, 16\}$ provides negligible improvement over uniform 4-bit for six of seven models. The sole exception is Qwen2.5-7B, where a cardinality-constrained solver achieves 10.91 perplexity at $4\times$ compression—a 29.4% reduction over uniform 4-bit (15.45)—while unconstrained solvers degrade catastrophically (32–37 PPL) due to 2-bit distortion compounding through the residual stream. The benefit depends not on GQA ratio or model scale, but on whether sensitivity heterogeneity is *spread* across multiple heads with cross-bitwidth gradients—a pattern observed in only one model ($N{=}1$) in our study. On the remaining six models—including Qwen2.5-14B, where scaling dilutes the spread pattern—non-uniform allocation provides no meaningful benefit: Yi-1.5-9B has the highest concentration tested (0.71, exceeding Qwen-7B's 0.61) yet gains nothing, because its single-outlier "spike" has no cross-bitwidth gradient. The cardinality constraint prevents catastrophic over-allocation on Qwen-7B and moderate degradation on flat models (Gemma-2: unconstrained $1.47\times$ degradation reduced to within 0.5% of uniform), while rendering solver choice irrelevant—constrained greedy and MCKP produce identical allocations—enabling a simple $O(H \log H)$ greedy algorithm. We characterize sensitivity heterogeneity across all seven models using concentration and distribution shape ($t_1/t_5$ ratio), identifying three patterns (spread, spike, flat) that correctly predict allocation benefit (7/7); the thresholds are preliminary, as only one model exhibits the spread pattern.

## 1 Introduction

Per-head mixed-precision quantization of the KV cache is a natural strategy for reducing memory during LLM inference: assign fewer bits to insensitive heads and more bits to sensitive ones, achieving lower perplexity than uniform quantization at the same average bit-width. Recent work on KV cache quantization explores both uniform schemes (Liu et al., 2024; Hooper et al., 2024; Lin et al., 2025; He et al., 2024) and token-level mixed-precision approaches (Ge et al., 2024; Lei and Ilager, 2026), while complementary methods reduce cache size through eviction (Zhang et al., 2023; Li et al., 2024). RateQuant (Zuo et al., 2026) formalizes per-head allocation as a rate-distortion optimization and shows that greedy allocation is optimal when bit-widths are consecutive integers $\{2, 3, 4, 5, 6\}$, exploiting the polymatroid structure of the feasible set (their Proposition 6).

In practice, hardware-efficient inference requires bit-widths drawn from $\{2, 4, 8, 16\}$—the formats natively supported by modern accelerators. On this non-consecutive set, the polymatroid structure breaks down, and greedy is no longer provably optimal. One might expect that exact MCKP solvers would fill this gap. They do not. On Qwen2.5-7B at $4\times$ compression, unconstrained greedy achieves 32.45 PPL and unconstrained MCKP achieves 36.92—both *far worse* than uniform 4-bit (15.45). The optimization does not merely underperform; it fails catastrophically.

The issue traces not to solver suboptimality but to a violation of the additive distortion assumption. Both solvers assign 68–75 of 112 KV heads to 2-bit, where per-head quantization errors compound through the residual stream. The additive model predicts $\sim$11 PPL; the actual PPL is 3–$4\times$ higher, because 2-bit distortions are orders of magnitude larger than 4-bit and compound superlinearly when many heads are quantized simulta-

neously.

This observation points to a minimal intervention: cap the number of 2-bit assignments. We add a single cardinality constraint $\sum_h \mathbb{1}[b_h{=}2] \leq k$ to the standard MCKP formulation (§3). On Qwen2.5-7B, setting $k{=}8$ (7% of heads) yields 10.91 PPL—a 29.4% reduction over uniform 4-bit. The constraint works by forcing the solver to use 2-bit only on the least-sensitive heads, while redistributing the freed budget to protect sensitive heads at 8-bit or 16-bit. This is not a heuristic patch: the constraint directly addresses the identified failure mode and is compatible with any solver. The constraint is beneficial even on flat-distribution architectures: on Gemma-2-9B, unconstrained MCKP assigns 31.5% of heads to 2-bit, producing a moderate $1.47\times$ degradation; the constraint reduces this to within 0.5% of uniform.

A surprising consequence is that the cardinality constraint renders solver choice irrelevant. Constrained greedy and MCKP produce identical allocations across all tested architectures (112/112 heads on Qwen, 256/256 on Llama, 255/256 on Mistral), enabling practitioners to use the simpler $O(H \log H)$ greedy algorithm. A per-layer brute-force oracle confirms near-optimality: the maximum gap is 0.02 PPL, 0.4% of the total improvement (§3.5).

However, non-uniform allocation does not help universally. On the remaining six models—including Qwen2.5-14B, where the $3.4\times$ increase in head count dilutes spread into flat—constrained MCKP provides no meaningful improvement. The difference traces to the *distribution shape* of head sensitivity, not GQA ratio or model scale alone: Qwen-7B's Layer-0 heads form a cluster of high sensitivity across multiple bitwidths ("spread"); Yi-1.5 and Llama have dominant outliers ("spike"); Gemma-2, Mistral, Qwen-14B, and Pythia show near-uniform or diluted sensitivity ("flat").

This motivates a question prior work does not address: *when* should a practitioner use per-head mixed-precision quantization? We propose a joint empirical decision rule based on concentration and distribution shape (§5.1) that correctly classifies all seven evaluated models (7/7), including a within-family scale contrast. Yi-1.5 is a critical test: its concentration (0.71) exceeds Qwen-7B's, but its single-outlier spike has no cross-bitwidth gradient, so MCKP cannot redistribute budget.

Our contributions are:

1. We identify 2-bit distortion compounding as a major contributing factor in the degradation observed under unconstrained per-head KV cache quantization on hardware-aligned bit-widths, and show that a single cardinality constraint on 2-bit assignments suffices to prevent it, reducing Qwen2.5-7B PPL from 32–37 (unconstrained) to 10.91 (constrained) at $4\times$ compression (§3.3, §4).
2. RateQuant (Zuo et al., 2026) formalized per-head MCKP and proved greedy optimality for consecutive integer bit-widths. Our algorithmic contribution is a single cardinality constraint (Eq. 2); the scientific contribution is the systematic finding that this constraint suffices to prevent compounding, yet six of seven tested models do not benefit from non-uniform allocation—the constraint renders solver choice irrelevant, with constrained greedy and MCKP producing equivalent allocations across all tested architectures (§3.5).
3. We provide an empirical characterization of head sensitivity heterogeneity across seven models, revealing three distribution shapes (spread, spike, flat) that predict allocation benefit. Concentration and $t_1/t_5$ ratio correctly classify all seven models (7/7), though the spread pattern is observed in only one model ($N{=}1$, Qwen2.5-7B); the suggested thresholds should be treated as preliminary (§5.1).

## 2 Related Work

**KV Cache Quantization.** Uniform-precision methods quantize the KV cache at a single bit-width across all heads, using techniques such as asymmetric per-channel/per-token schemes (Liu et al., 2024; Hooper et al., 2024), quantization–serving co-design (Lin et al., 2025), salient-token identification (He et al., 2024), low-rank plus sparse correction (Kang et al., 2024), sliding-window adaptation (Duanmu et al., 2024), rotation-based outlier smoothing for robust 2-bit (Su et al., 2025), 2-bit with adaptive token retention (Sharma et al., 2025), and adapter-based inter-layer prediction at 2–2.5 bits (Shutova et al., 2025). We extend the design space to per-head non-uniform allocation over hardware-aligned bit-widths $\{2, 4, 8, 16\}$. Our per-head bit-width allocation is orthogonal to per-channel quantization

schemes such as KIVI: KIVI determines *how* each head is quantized (asymmetric per-channel scaling), while our method determines *at what precision* (bit-width selection across heads). The two can be composed, though the per-channel method dominates when available (§4.2); per-head allocation provides independent value in symmetric-only environments where per-channel asymmetric quantization is unavailable.

**KV Cache Eviction.** Complementary to quantization, eviction methods reduce cache size by discarding tokens—retaining heavy-hitters (Zhang et al., 2023), per-head important tokens (Li et al., 2024; Ge et al., 2024), persistent-importance tokens (Liu et al., 2023), attention sinks (Xiao et al., 2024), or progressively fewer tokens in deeper layers (Yang et al., 2024). Recent per-head eviction methods allocate non-uniform budgets across heads based on retrieval-vs-reasoning importance (Fu et al., 2025) or adaptive optimization extending SnapKV/PyramidKV (Feng et al., 2025), confirming that head-level heterogeneity is exploitable for eviction; our work shows it is equally exploitable for quantization.

**Mixed-Precision Quantization (Weights).** Mixed-precision allocation has a long history at the weight level. HAWQ (Dong et al., 2019) and HAWQ-V2 (Dong et al., 2020) use second-order (Hessian-based) sensitivity to assign per-layer bit-widths, with V2 introducing block-wise trace-weighted scoring and Pareto-frontier selection; HAQ (Wang et al., 2019) learns an allocation by reinforcement learning under hardware-aware latency/energy constraints. These methods operate on weights, where per-layer distortions are approximately additive and sensitivity varies smoothly across layers. KV cache quantization presents a qualitatively different challenge: 2-bit head distortions compound super-additively through the residual stream (§3.3), a phenomenon absent in weight quantization where each layer's parameters are independently quantized. Our primary contribution is not the allocation algorithm itself—any MCKP or greedy solver suffices—but the diagnosis and fix of this compounding effect via cardinality constraints, and the characterization of when non-uniform allocation is beneficial (§5.1).

**Mixed-Precision Rate-Distortion Allocation.** RateQuant (Zuo et al., 2026) formulates per-head bit-width selection as an MCKP and proves greedy optimality for consecutive bit-widths $\{2, \ldots, 6\}$ via the polymatroid structure, which relies on the assumption that per-head distortions decompose additively into the global objective—valid when adjacent bit-widths differ by a single bit so that quantization noise stays in the small-perturbation regime. On hardware-aligned $\{2, 4, 8, 16\}$, the $4\times$ gap between 2-bit and 4-bit drives 2-bit distortion to compound super-additively, breaking the polymatroid structure and the additivity assumption with it. Concretely, running RateQuant's greedy algorithm (Algorithm 1) directly on $\{2, 4, 8, 16\}$ yields 32.45 PPL on Qwen2.5-7B at $4\times$ compression—catastrophically worse than uniform 4-bit (15.45)—whereas our cardinality constraint on simultaneously active 2-bit heads recovers 10.91 PPL at the same average bit-width. The constraint is the minimal fix that restores the additive regime in which RateQuant's polymatroid argument applies. Related mixed-precision KV-cache work operates at coarser granularity—layer-level via gradient-based sensitivity (Li et al., 2025a) or Pareto search over hardware-friendly precision pairs (Li et al., 2025b), block-level via progressive decoding-step quantization for long-CoT reasoning (Liu et al., 2025), channel-level via dynamic precision boost on sensitive Key channels (Xia et al., 2025), token-level via a learned on-device controller (Haeri Boroujeni et al., 2026), memory-budget-level (Lei and Ilager, 2026)—at the weight level (Kim et al., 2024), or dynamically at inference time (Zhang et al., 2025); our allocation is static and per-head.

**Attention Head Analysis.** Prior work studies head redundancy and pruning (Voita et al., 2019; Michel et al., 2019), focusing on which heads can be removed entirely. Our analysis asks a different question: which heads are sensitive to *quantization*, and how heterogeneous that sensitivity is across architectures—a property that determines whether non-uniform allocation is beneficial (§5.1).

## 3 Method

### 3.1 Problem Formulation

We formulate per-head KV cache quantization as a constrained optimization problem. Let $h \in \{1, \ldots, H\}$ index the KV heads across all layers, $b_h \in \mathcal{B}$ the bit-width assigned to head $h$, and

$\mathcal{B} = \{2, 4, 8, 16\}$ the set of hardware-aligned bit-widths. Each head $h$ has a profiled distortion function $d_h(b)$, the PPL increase when head $h$ is quantized to $b$ bits. Given a memory budget ratio $R$ (e.g., $R = 0.25$ for $4\times$ compression), we seek the allocation that minimizes total distortion:

$$\min_{\{b_h\}} \sum_{h=1}^{H} d_h(b_h) \quad \text{s.t.} \quad \frac{1}{H \cdot 16} \sum_{h=1}^{H} b_h \leq R. \quad (1)$$

This is a multiple-choice knapsack problem (MCKP): each head selects one bit-width from $\mathcal{B}$ to minimize total distortion under a budget. RateQuant (Zuo et al., 2026) solves this for consecutive bit-widths $\{2, 3, 4, 5, 6\}$, where the polymatroid structure of the feasible set guarantees that greedy allocation is optimal (their Proposition 6). Hardware-aligned bit-widths $\{2, 4, 8, 16\}$ are non-consecutive; the gaps break this structure, and greedy is no longer guaranteed to match the MCKP optimum. Moreover, as we show in §3.3, the objective itself is not well-behaved: distortions from 2-bit heads compound non-additively, making unconstrained optimization unreliable.

### 3.2 PPL-based Per-Head Profiling

We use symmetric per-channel quantization with per-head granularity: each KV head is independently quantized to one of the candidate bit-widths $\mathcal{B} = \{2, 4, 8, 16\}$, without per-token scaling. We profile each head by quantizing it to each $b \in \mathcal{B}$ while keeping all other heads at bf16, and measuring the resulting PPL increase $d_h(b)$ on a calibration set of 64 sequences $\times$ 2048 tokens from WikiText-2 train. This yields a distortion table of $H \times |\mathcal{B}|$ entries that serves as input to the solver.

The optimization in Eq. (1) relies on an additive distortion assumption: the total PPL degradation from a mixed-precision allocation is approximately $\sum_h d_h(b_h)$. This holds when per-head distortions are small and independent—a reasonable approximation for 4-bit and 8-bit quantization, where individual head distortions are typically below 0.1 PPL. At 2-bit, however, individual distortions are orders of magnitude larger, and the additive assumption breaks down.

### 3.3 Compounding Effect

When many heads are simultaneously quantized to 2-bit, per-head distortions compound non-additively: truncation errors from multiple 2-bit heads accumulate through the residual stream, and once their count exceeds the network's capacity to compensate, PPL degrades super-linearly.

Unconstrained solvers are blind to this effect because the additive objective underestimates the cost of large-scale 2-bit assignment. On Qwen2.5-7B, unconstrained greedy assigns 68 heads to 2-bit and achieves 32.45 PPL; unconstrained MCKP assigns 75 heads and achieves 36.92—both far worse than uniform 4-bit (15.45). Standard-resolution profiling detects non-zero $d_h(2)$ for only 37 of 112 heads; the remaining 75 heads have $d_h(2) \approx 0$, so most 2-bit assignments appear free to the additive objective. The resulting additive prediction is $\sum_h d_h(b_h) + \text{baseline} \approx 11$ PPL—3–4$\times$ lower than the actual 32–37 PPL, because the compounding interaction between simultaneously quantized heads is invisible to per-head profiling.

This compounding is specific to 2-bit quantization and is *super-additive*: quantizing the top-$k$ most sensitive heads to 2-bit simultaneously produces PPL $1.56\times$ the sum of their individual distortions at $k$=8, growing to $4.89\times$ at $k$=64. A random-$k$ control confirms that sensitivity ordering matters: at $k$=64, random selection yields 55.94 PPL vs. 22.95 for top-$k$, because random draws are more likely to include high-distortion heads that amplify compounding. At 4-bit and above, per-head distortions are small enough that additivity holds even when all heads share the same bit-width (uniform 4-bit achieves 15.45 PPL, close to the sum-of-distortions prediction). The qualitative difference arises because 2-bit quantization has far larger per-head distortion, saturating the error tolerance of the residual stream.

The relationship between $k$ and PPL is U-shaped (Figure 2, Appendix A): at $k$=0 (uniform 4-bit), PPL is 15.45; allowing a few 2-bit heads reduces PPL by freeing budget for 8-bit heads on sensitive positions, reaching a minimum of 10.89 at $k$=8; beyond $k$=8, compounding drives PPL upward.

### 3.4 Cardinality Constraint

The compounding effect motivates a simple addition to Eq. (1): a cardinality constraint on 2-bit assignments.

$$\sum_{h=1}^{H} \mathbb{1}[b_h = 2] \leq k. \quad (2)$$

The constraint parameter $k$ is selected by a

Table 1: Solver comparison on Qwen2.5-7B at $4\times$ compression. Unconstrained solvers over-allocate 2-bit heads and fail catastrophically. The cardinality constraint ($k \leq 8$) makes greedy and MCKP produce identical allocations.[1]

| Solver | Constraint | PPL | Bit distribution | | |
|---|---|---|---|---|---|
| | | | 2-bit | 4-bit | 8/16-bit |
| Greedy | None | 32.45 | 68 | 30 | 14 |
| MCKP | None | 36.92 | 75 | 33 | 4 |
| Greedy | $k \leq 8$ | 10.91 | 8 | 100 | 4 |
| MCKP | $k \leq 8$ | 10.91 | 8 | 100 | 4 |
| Uniform | — | 15.45 | 0 | 112 | 0 |

small-scale sweep on calibration data: solve the constrained problem for each candidate $k$, evaluate calibration PPL, and pick the value at the minimum of the U-shaped curve (Figure 2). For Qwen2.5-7B with 112 KV heads, $k{=}8$ ($\sim$7% of heads).

The constraint is a single linear inequality, compatible with any solver—greedy, MCKP, or integer linear programming—and adds negligible overhead to the optimization.

One might expect Layer-0 heads to require explicit protection. On Qwen2.5-7B, however, the profiler assigns very high 2-bit distortion to Layer-0 heads (L0H0: $+31.9\%$, L0H2: $+23.2\%$, L0H3: $+11.7\%$ relative to bf16), so no solver would assign them 2-bit under any reasonable budget. Running the solver with and without manual Layer-0 exclusion produces identical allocations; no manual rule is needed.

**Restoring additivity.** The additive objective treats per-head distortions as independent, but 2-bit compounding violates this once many 2-bit heads coexist. Restricting 2-bit heads to $k \ll H$ pushes the allocation back into a near-additive regime where the profiler's prediction agrees with measured PPL. For Qwen2.5-7B, $k{=}8$ ($\sim$7% of heads) is few enough that compounding contributes negligibly, consistent with the U-shaped curve minimum (Figure 2). This explains the constrained–unconstrained gap in Table 1: constrained greedy operates inside the additive regime (10.91 PPL, matching the $\approx$11 additive estimate), while unconstrained solvers extrapolate outside it and miss true PPL by 3–4$\times$.

### 3.5 Solver Equivalence

With the cardinality constraint, greedy and MCKP produce identical results (Table 1): same PPL (10.91), same bit distribution on Qwen2.5-7B.

**Empirical observation.** When $k \ll H$ and the top-$k$ heads ranked by marginal benefit $d_h(4) - d_h(2)$ contain no ties, constrained greedy and MCKP produce identical per-head allocations. The constraint restricts the feasible set so that the only decision is which $k$ heads to assign 2-bit; the marginal-benefit ordering determines this uniquely, so greedy's local choice coincides with the global optimum. We verify this across three architectures: Qwen2.5-7B (112/112 heads identical), Llama-3.1-8B (256/256), Mistral-7B-v0.3 (255/256; the single difference is a zero-distortion tie with no PPL impact). Without the constraint, greedy cascades into excessive 2-bit allocation, as each additional 2-bit head looks locally beneficial but contributes to compounding. Practitioners can therefore use the simpler $O(H \log H)$ greedy algorithm without sacrificing optimality.

A per-layer brute-force oracle on Qwen2.5-7B confirms near-optimality: of 8 layers with non-trivial allocations, MCKP matches the oracle on 5; the remaining 3 have a maximum gap of 0.018 PPL (0.4% of the total improvement), closing the optimality chain from greedy through MCKP to brute-force.

## 4 Experiments

### 4.1 Experimental Setup

**Models.** We evaluate seven models spanning five architecture families and diverse attention configurations: **Qwen2.5-7B** (GQA 7:1, 28 layers $\times$ 4 KV heads = 112), **Qwen2.5-14B** (GQA 5:1, 48 $\times$ 8 = 384), **Yi-1.5-9B** (GQA 8:1, 48 $\times$ 4 = 192), **Llama-3.1-8B** (GQA 4:1, 32 $\times$ 8 = 256), **Mistral-7B-v0.3** (GQA 4:1, 32 $\times$ 8 = 256, sliding window attention), **Gemma-2-9B** (GQA 2:1, 42 $\times$ 8 = 336, `head_dim`=256, attention logit soft-capping), and **Pythia-1.4B** (MHA 1:1, 24 $\times$ 16 = 384). This selection spans GQA ratios from 1:1 (MHA) to 8:1 and covers standard, sliding-window, softcapped, and multi-head attention.

**Evaluation.** We report perplexity (PPL, $\downarrow$) on the WikiText-2 test split (298,938 tokens) under bf16 inference with stride 512. For downstream

[1] Constrained results use high-resolution profiling; unconstrained rows use standard-resolution profiling to illustrate the failure mode (the exact PPL is irrelevant—any unconstrained allocation fails catastrophically).

Table 2: PPL (↓) on WikiText-2 at 4× compression. Unconstrained failure: catastrophic on Qwen-7B (3.3× bf16), severe on Pythia (1.9×), moderate on Gemma-2 (1.5×), mild on remaining incl. Qwen-14B (∼1.1×). Greedy and MCKP columns are both unconstrained; identical allocations on 5/7 models (equivalence verified experimentally on Qwen-7B and Qwen-14B; Qwen-14B and Pythia Greedy values inferred via this equivalence). Best constrained result per model in **bold**. *BOS-corrected.

| Model | Baseline | Uniform | Unconstrained | | Ours |
|---|---|---|---|---|---|
| | (bf16) | (4-bit) | Greedy | MCKP | (Constr.) |
| Qwen2.5-7B | 9.81 | 15.45 | 32.45 | 36.92 | **10.91** |
| Qwen2.5-14B | 6.87 | 7.20 | 7.69 | 7.69 | 7.22 |
| Yi-1.5-9B | 7.02 | 7.22 | 7.56 | 7.56 | **7.22** |
| Llama-3.1-8B | 8.09 | 8.47 | 8.63 | 8.63 | **8.45** |
| Mistral-7B-v0.3 | 7.04 | 7.13 | 7.57 | 7.60 | 7.13 |
| Gemma-2-9B* | 6.03 | 6.17 | 8.85 | 8.85 | 6.20 |
| Pythia-1.4B | 16.49 | 24.29 | 31.54 | 31.54 | **24.15** |

evaluation, we use ARC-Challenge, HellaSwag, PIQA, and WinoGrande under 0-shot prompting.

**Profiling.** Per-head sensitivity is measured by independently quantizing each head to each candidate bit-width in $\{2, 4, 8, 16\}$ and recording the PPL increase over the bf16 baseline. All results use high-resolution profiling (64 sequences $\times$ 2048 tokens from WikiText-2 train); Appendix I compares against standard-resolution profiling (16 seq $\times$ 512 tok).

**Compression and constraint.** The default target is 4× compression (average 4 bits, budget ratio 0.25). The cardinality constraint limits 2-bit assignments to at most $k$ heads, where $k$ is selected via the compounding sweep described in §3 ($k$=8 for Qwen-7B and Qwen-14B, $k$=2 for Yi-1.5 and Llama, $k$=2 for Mistral, $k$=8 for Gemma-2 and Pythia).

**Baselines.** We compare against: (1) uniform 4-bit quantization; (2) unconstrained greedy allocation (RateQuant-style (Zuo et al., 2026)); (3) unconstrained MCKP allocation; (4) an L0-random heuristic that assigns Layer-0 heads to 8-bit and $k$ random non-Layer-0 heads to 2-bit, with remaining heads at 4-bit (5 seeds; we report mean±std and median); and (5) a naive positional heuristic that assigns Layer-0 to 8-bit and the $k$ deepest heads to 2-bit.

## 4.2 Main Results

On Qwen2.5-7B, constrained MCKP achieves 10.91 PPL, a 29.4% reduction over uniform 4-bit (15.45). Without the constraint, both greedy (32.45) and MCKP (36.92) perform *worse* than uniform—a catastrophic failure from excessive 2-bit assignments compounding across layers. Across 3 calibration seeds, constrained MCKP yields mean PPL 10.99±0.06 with 90.2% three-way allocation agreement, confirming profiling stability.

On Llama-3.1-8B, the result is essentially indistinguishable from uniform (8.45 vs. 8.47, $\Delta < 0.02$ PPL). Hires profiling reveals concentration 0.57 (comparable to Qwen's 0.61) but a spike pattern ($t_1/t_5$=0.85) dominated by a single head (L2H5), leaving no cross-bitwidth gradient to exploit.

On Yi-1.5-9B, constrained MCKP ($k$=2) matches uniform (7.22); despite the highest concentration tested (0.71), sensitivity is dominated by a single outlier (L4H3, $t_1/t_5$=0.90), a spike pattern with no cross-bitwidth gradient. On Mistral-7B-v0.3, constrained MCKP ($k$=2) matches uniform (7.13 vs. 7.13); near-zero sensitivity differences and sliding-window attention leave no room for reallocation gains.

On Gemma-2-9B* (GQA 2:1), constrained MCKP ($k$=8) achieves 6.20 PPL vs. uniform 6.17 (+0.4%). Despite flat sensitivity (concentration 0.23), unconstrained MCKP degrades to 8.85 (1.47× bf16) from 31.5% 2-bit over-allocation; the constraint reduces this to 8 heads, confirming universal over-allocation prevention.

Unconstrained failure severity tracks heterogeneity shape: *catastrophic* on Qwen-7B (3.3× bf16, 60% heads at 2-bit), *severe* on Pythia (1.9×), *moderate* on Gemma-2 (1.47×), and *mild* on remaining models (∼1.07–1.12×) including Qwen-14B (1.12×, 20% 2-bit)—the within-family contrast directly reflects shape (spread vs. flat). See Appendix C (allocation visualization) and Appendix F (budget sweep) for additional detail.

MCKP consistently outperforms random L0-constrained allocation and naive positional baselines across all three tested models (median gap 5–16%; Table 5 in Appendix B).

**Comparison with KIVI.** KIVI (Liu et al., 2024) uses asymmetric per-channel quantization with per-token scaling, applied uniformly across all heads. At 4-bit ($r$=0), KIVI achieves near-lossless

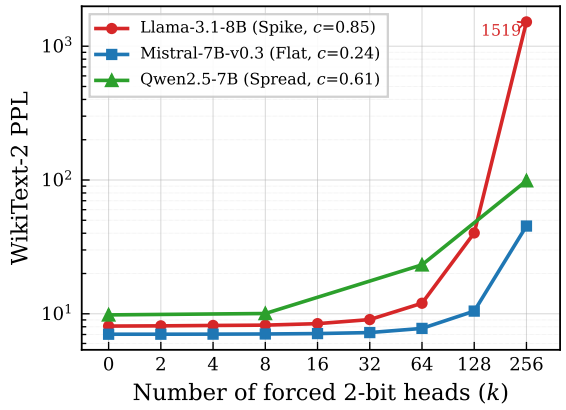



Figure 1: Compounding effect across architectures. As the number of 2-bit heads ($k$) increases, PPL degradation follows architecture-dependent trajectories: Llama (spike, $c$=0.57) explodes to 1519× at $k$=256 due to a single extreme-sensitivity head compounding through the residual stream; Qwen (spread, $c$=0.61) degrades 10×; Mistral (flat, $c$=0.24) remains most resilient (6.4×). All three stay near-baseline for $k \leq 8$, validating the cardinality constraint regime. Qwen saturates at $k \approx 96$ due to its 112 KV heads (maximum ∼96 eligible for 2-bit assignment).

PPL (9.91 on Qwen, 8.14 on Llama vs. 9.81/8.09 bf16), while our symmetric scheme yields 15.45 and 8.47—a gap from the quantization method, not allocation. Within the symmetric scheme, constrained MCKP reduces Qwen from 15.45 to 10.91 (−4.54 PPL) by reallocating bits across heads; on Llama (spike), the gain is negligible (−0.02). The two approaches are orthogonal and composable: KIVI selects the per-channel scheme, our method selects the per-head bit-width. However, composability does not imply additive benefit: on Qwen, KIVI+MCKP achieves 9.90 PPL—identical to KIVI alone (9.90; MCKP alone: 10.89), as the per-channel method subsumes the head-level reallocation signal. Per-head allocation provides independent value in symmetric-only deployment environments—common in hardware accelerators that lack per-channel asymmetric support—where it is the sole degree of freedom for exploiting sensitivity heterogeneity.

### 4.3 Cross-Architecture Compounding

Figure 1 shows the compounding sweep across three architectures representing the three sensitivity patterns. At small $k$ ($\leq 8$), all models tolerate 2-bit assignments with minimal PPL increase ($\leq 2.5\%$ over baseline), confirming that the cardinality constraint ($k$=2–8) operates in the near-additive regime regardless of architecture.

Beyond $k$=8, divergence is stark. Llama (spike pattern) remains stable through $k$=32 (PPL 9.05, 1.12× baseline) but explodes at $k$=128 (40.21) and $k$=256 (1519, 188× baseline)—the single extreme-sensitivity head (L2H5) amplifies compounding once enough surrounding heads are also degraded. Qwen (spread pattern) rises earlier, reaching 23.22 at $k$=64 (2.4×) and 98.75 at $k$=256 (10.1×), as its cluster of Layer-0 sensitive heads compound non-additively at moderate $k$. Mistral (flat pattern) is the most resilient: PPL stays below 8 through $k$=64 and reaches only 45.28 at $k$=256 (6.4×), consistent with its near-uniform sensitivity distribution offering no single failure mode.

These curves directly motivate architecture-specific $k$ selection: the compounding onset ($k$ at which PPL exceeds 1.5× baseline) is ∼32 for Qwen, ∼64 for Llama, and ∼128 for Mistral, with the optimal constrained $k$ lying well below each onset.

Table 3: 0-shot accuracy change (Ours − Uniform, pts) at 4× compression. Qwen values are mean over 3 calibration seeds ($\pm 1\sigma$). Qwen shows gains on HS/PIQA/WG; the remaining models show near-neutral change, tracking PPL results.

| Model | ARC-C | HS | PIQA | WG | Avg |
|---|---|---|---|---|---|
| Qwen2.5-7B | +0.0±0.4 | +4.1±0.2 | +2.0±0.2 | +1.6±1.2 | +1.9 |
| Yi-1.5-9B | −2.1 | −0.4 | +0.0 | +0.3 | −0.5 |
| Llama-3.1-8B | −0.8 | +0.0 | −0.1 | +0.5 | −0.1 |
| Mistral-7B | +0.0 | +0.0 | −0.1 | −0.1 | +0.0 |
| Gemma-2-9B | −0.1 | +0.0 | +0.4 | +0.6 | +0.2 |

### 4.4 Downstream Evaluation

On Qwen, constrained MCKP improves average 0-shot accuracy by 1.9 pts over uniform (mean over 3 seeds, $\pm 1\sigma$ in Table 3), with the largest gains on HellaSwag (+4.1) and PIQA (+2.0). The four non-spread models show near-neutral average change (Yi −0.5, Llama −0.1, Mistral +0.0, Gemma-2 +0.2 pts); Mistral is effectively unchanged across all tasks, consistent with its flat sensitivity profile. ARC-Challenge shows the largest per-task sensitivity across all models (Yi −2.1, Llama −0.8, Gemma-2 −0.1 pts), consistent with multi-step reasoning where KV cache errors accumulate across longer inference chains. Cross-dataset (Appendix G, >96% bit-assignment agreement across WikiText-2/C4) and long-context (Appendix L, Qwen uniform 11.15×

bf16 at 8K vs. MCKP 1.23×) analyses confirm allocation robustness.

## 5 Analysis

### 5.1 Heterogeneity Characterization

The benefit of non-uniform allocation depends on how unevenly head sensitivities are distributed. We quantify this with a *top-5% concentration metric*: sort per-head 2-bit distortions $d_h(2)$ in descending order and compute the fraction of total distortion contributed by the top 5% of heads. High concentration indicates that a few heads are disproportionately sensitive; low concentration indicates near-uniform sensitivity.

Table 4 summarizes the results across seven models at $4\times$ compression. The central finding is that MCKP benefit requires *spread heterogeneity*—multiple heads with cross-bitwidth sensitivity gradients—rather than sensitivity concentrated in a single outlier at one bitwidth.

Qwen2.5-7B exemplifies the spread pattern (top-5% concentration 0.61, $t_1/t_5$=0.57), with Layer-0 heads contributing the majority of total 2-bit distortion across multiple bitwidths. Constrained MCKP reduces degradation from +5.64 PPL (uniform) to +1.10 PPL, a 29.4% improvement over uniform 4-bit.

Yi-1.5-9B is a critical counterexample: despite the highest concentration tested (0.71, exceeding Qwen's 0.61), constrained MCKP matches uniform (+0.20 vs. +0.20). Yi-1.5's sensitivity is dominated by a single extreme outlier: Layer-4 Head-3 ($d_h(2)$=13.12, second-ranked head: 0.55, a 24× gap; $t_1/t_5$=0.90). Critically, L4H3's 4-bit and 8-bit deltas are $\approx 0$, so upgrading it from 2-bit to 4-bit removes the distortion, but budget freed by downgrading other heads has nowhere useful to go—a "spike" pattern with sensitivity at only one bitwidth. In contrast, Qwen's Layer-0 heads exhibit non-zero deltas at multiple bitwidths (e.g., L0H0: 2-bit delta =6.74, with positive 4-bit and 8-bit deltas), giving the solver a multi-bitwidth gradient across several heads—a "spread" pattern enabling meaningful reallocation.

Llama-3.1-8B has concentration 0.57—comparable to Qwen's 0.61—yet its spike pattern ($t_1/t_5$=0.85), dominated by a single head (L2H5), yields allocation indistinguishable from uniform (+0.36 vs. +0.38), demonstrating that concentration alone is insufficient without spread structure. Mistral-7B-v0.3 ($t_1/t_5$=0.41) and Gemma-2-9B ($t_1/t_5$=0.38) both exhibit flat distributions with low concentration (0.24 and 0.23), leaving no reliable signal; constrained MCKP is slightly worse than uniform in both cases (+0.13 vs. +0.10 for Mistral; +0.16 vs. +0.14 for Gemma-2).

Table 4: Head sensitivity heterogeneity and constrained allocation benefit at $4\times$ compression. $t_1/t_5$: ratio of the most sensitive head's 2-bit distortion to the sum of the top-5% of heads' distortions (low=spread, high=spike); the percentage threshold ensures comparability across models with different head counts (112–384). *BOS-corrected.

| Model | GQA Ratio | Top-5% Conc. | $t_1/t_5$ | Δ PPL (Uniform) | Δ PPL (Ours) | Shape |
|---|---|---|---|---|---|---|
| Qwen2.5-7B | 7:1 | 0.61 | 0.57 | +5.64 | +1.10 | Spread |
| Qwen2.5-14B | 5:1 | 0.33 | 0.68 | +0.33 | +0.35 | Flat |
| Yi-1.5-9B | 8:1 | 0.71 | 0.90 | +0.20 | +0.20 | Spike |
| Llama-3.1-8B | 4:1 | 0.57 | 0.85 | +0.38 | +0.36 | Spike |
| Mistral-7B-v0.3 | 4:1 | 0.24 | 0.41 | +0.10 | +0.10 | Flat |
| Gemma-2-9B* | 2:1 | 0.23 | 0.38 | +0.14 | +0.16 | Flat |
| Pythia-1.4B | 1:1 | 0.24 | 0.28 | +7.80 | +7.96 | Flat |

Pythia-1.4B extends the evaluation to MHA (1:1 ratio, 384 heads). Low concentration (0.24) and $t_1/t_5$=0.28 place it in the flat category; constrained MCKP confirms this (+7.96 vs. uniform +7.80, +0.7%).

**Scale-dependent concentration dilution.** Qwen2.5-14B (GQA 5:1, 384 heads) provides a within-family comparison: $t_1/t_5$ is similar (0.57 vs. 0.68), but concentration drops from 0.61 (7B) to 0.33 (14B)—diluted by 3.4× more heads into the flat regime. Sensitivity is dominated by a single head (L0H5, $d_h(2)$=1.845; second-ranked: 0.237, a 7.8× gap), but 384 heads dilute cumulative signal below the concentration threshold. Concentration, not within-head structure, is thus the primary discriminator: the rule correctly identifies Qwen-7B as the single model×scale combination where non-uniform allocation helps.

The per-head heatmap (Figure 4, Appendix D) confirms this visually: Qwen's Layer-0 heads show high distortion while mid-to-deep layers are near-zero (spread); Yi-1.5 shows a single bright spot at Layer-4 Head-3 (64% of total delta) with the remaining 191 heads near-uniform (spike); Gemma-2 and Mistral show uniformly low distortion.

### 5.2 Empirical Decision Rule

MCKP benefit requires not just high concentration but a specific *shape*—spread heterogeneity across

multiple heads with cross-bitwidth sensitivity gradients. This motivates a joint empirical decision rule for deciding when non-uniform allocation is worthwhile. Given the profiling results from §3.2, compute the top-5% concentration and $t_1/t_5$ ratio, then select:

- **Spread** (concentration $\geq 0.50$ *and* $t_1/t_5 \leq 0.70$): non-uniform recommended. Qwen-7B (0.61, $t_1/t_5$=0.57): $-29.4\%$ PPL.
- **Spike** ($t_1/t_5 > 0.70$ *and* concentration $\geq 0.25$): use uniform. A dominant outlier with no cross-bitwidth gradient cannot be exploited. Yi-1.5 (0.71, 0.90) and Llama (0.57, 0.85): no benefit.
- **Flat** (concentration $< 0.25$, or concentration $\in [0.25, 0.50)$ with $t_1/t_5 \leq 0.70$): use uniform. Too little cumulative delta for meaningful reallocation. Mistral (0.24), Gemma-2 (0.23), Qwen-14B (0.33), and Pythia (0.24): no benefit or counterproductive.

Both conditions are needed: Llama (0.57) and Yi-1.5 (0.71) both exceed the concentration threshold yet show no MCKP benefit; their spike shapes ($t_1/t_5 \geq 0.70$) correctly override. A concentration-only rule would misclassify both as spread; $t_1/t_5$ is the critical second discriminator. The rule correctly classifies all seven evaluated models (7/7), including the within-family scale contrast (Qwen-7B spread vs. 14B flat) and the MHA architecture (Pythia-1.4B flat). Only Qwen-7B exhibits the spread pattern ($N$=1); the suggested thresholds should be treated as preliminary until confirmed on additional models. The rule reuses profiling data already required by the solver and adds no overhead.

**Why is Qwen-7B the only spread model?** We offer two non-exclusive hypotheses, both requiring further validation. First, Qwen-7B's high GQA ratio (7:1) means each KV head aggregates attention from 7 query heads, potentially driving greater functional specialization and thus larger inter-head sensitivity differences. However, Yi-1.5 (8:1) and Qwen-14B (5:1) have comparable or higher GQA ratios yet do not exhibit spread, so GQA ratio alone is insufficient. Second, scale-dependent concentration dilution may play a role: Qwen-14B (same architecture family, $3.4\times$ more heads) dilutes into the flat regime despite similar within-head structure ($t_1/t_5$ 0.68 vs. 0.57). This suggests that increasing head count may systematically reduce concentration, making spread less likely at larger scales—a hypothesis consistent with the Qwen-7B/14B contrast but requiring evaluation on additional model families and scales.

Additional analyses—granularity tradeoff (Appendix E), aggressive compression, profiling resolution and stability, per-layer optimality, and KV separation—are in Appendices E–J.

## 6 Conclusion

Per-head mixed-precision KV cache quantization over hardware-aligned bit-widths {2, 4, 8, 16} requires a cardinality constraint on 2-bit assignments to prevent non-additive distortion compounding. On Qwen2.5-7B, limiting 2-bit heads to $k \leq 8$ reduces PPL from 15.45 (uniform 4-bit) to 10.91 ($-29.4\%$) at $4\times$ compression, while making solver choice irrelevant: constrained greedy and MCKP produce identical allocations across all tested architectures. Whether non-uniform allocation helps depends on the *shape* of head sensitivity heterogeneity—spread patterns (Qwen-7B, $t_1/t_5$=0.57) yield large gains, while spike (Yi-1.5, Llama) and flat (Mistral, Gemma-2, Qwen-14B, Pythia) patterns show no benefit. An empirical characterization based on concentration and $t_1/t_5$ ratio correctly predicts allocation benefit for all seven evaluated models (7/7), including a within-family scale contrast where Qwen-14B's spread dilutes into flat. Extending to 70B+ models, production inference engines, and per-side K/V cardinality constraints (Appendix J) are natural next steps.

## Limitations

**$N$=1 spread.** The spread pattern is observed in only one model (Qwen2.5-7B); the decision rule thresholds (concentration $\geq 0.50$, $t_1/t_5 \leq 0.70$) are calibrated on seven models with a single positive example and should be treated as a preliminary observation rather than a validated classifier.

**Model scale.** The largest model evaluated is 14B parameters (384 heads). The Qwen-7B/14B contrast suggests that concentration dilutes with increasing head count, which may make spread less likely at larger scales; this trend is a hypothesis, not a confirmed finding.

**Runtime.** All latency measurements use PyTorch-level per-head `quantize`/`dequantize` dispatch and do

not represent optimized CUDA kernel performance. The overhead reported in Appendix K is an upper bound on the mixed-precision penalty; production inference engines with fused attention kernels would likely narrow this gap. No real-kernel latency benchmarks are reported.

**Downstream scope.** Downstream evaluation covers four 0-shot classification tasks (ARC-C, HellaSwag, PIQA, WinoGrande). Generation tasks (summarization, translation, long-form QA) are not evaluated, and no confidence intervals are reported for the current single-seed results.

**Calibration domain.** Cross-dataset stability is verified only between WikiText-2 and C4 (96–97% bit-assignment agreement); other domains (code, dialogue, multilingual) are untested.

**Long-context.** Long-context evaluation extends only to 8K tokens; deployment scenarios at 32K–128K context lengths are not tested.

**Missing direct competitors.** No existing tool performs per-head mixed-precision KV cache quantization on hardware-aligned bit-widths, precluding direct comparison with an established baseline at the same granularity.

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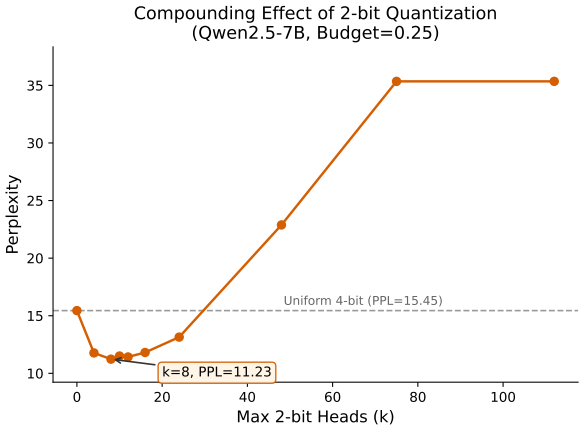



Figure 2: PPL on Qwen2.5-7B ($4\times$ compression) as a function of $k$, the maximum number of 2-bit heads. Too few 2-bit heads waste budget on 4-bit; too many trigger compounding. The minimum occurs at $k$=8.

Table 5: Constrained MCKP vs. heuristic baselines at $4\times$ compression. L0-random reports median and mean±std over 5 seeds. All methods use per-model optimal $k$ ($k$=8 for Qwen, $k$=2 for Llama, $k$=2 for Mistral). Llama and Mistral use hires profiling.

| Model | MCKP | L0-random | | Naive |
|---|---|---|---|---|
| | | Median | Mean±Std | Positional |
| Qwen2.5-7B | 11.23 | 11.52 | 11.69±0.47 | 12.23 |
| Llama-3.1-8B | 8.45 | 8.62 | 8.62±0.07 | 8.58 |
| Mistral-7B-v0.3 | 7.13 | 7.20 | 7.24±0.07 | 7.21 |

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## A Compounding Curve

## B Heuristic Baselines

On Qwen, MCKP (11.23) beats the L0-random median (11.52) by 0.29 PPL and the naive posi-

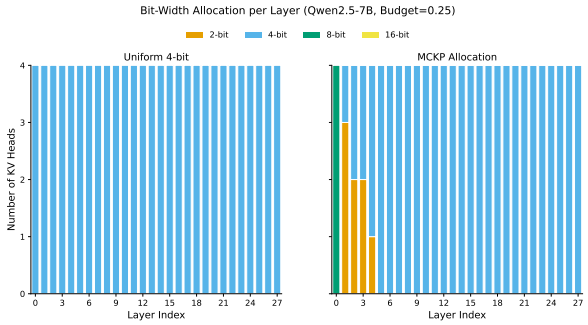



Figure 3: Bit-width allocation on Qwen2.5-7B at 4× compression. Uniform assigns all heads to 4-bit. Constrained MCKP protects Layer-0 heads at 8-bit and assigns the 8 least-sensitive heads to 2-bit.

tional heuristic (12.23) by 1.00 PPL. On Llama ($k$=2), all methods cluster within 0.17 PPL; on Mistral ($k$=2), MCKP (7.13) matches uniform (7.13), while heuristic baselines remain slightly worse. Heuristic baselines use a slightly higher budget (0.256 vs. 0.248, a 3% difference); despite this advantage, they still underperform MCKP.

## C Allocation Visualization

## D Per-Head Sensitivity Heatmap

## E Granularity Tradeoff

Quantization budget can in principle be allocated at three granularities, each with a distinct cost–benefit profile. *Layer-level* allocation (KVmix (Li et al., 2025a), KVTuner (Li et al., 2025b)) keeps the search space small ($L$ layers) but forces all heads in a layer to share one bitwidth. This is fundamentally limited when within-layer variance is large: in Qwen2.5-7B Layer-0, per-head 2-bit deltas are 6.74, 2.12, 1.31, and 1.24—a more than 5× spread that any layer-level assignment must average over. *Per-head* allocation (ours) exposes this heterogeneity to the solver and yields a 29.4% PPL improvement on Qwen, at the cost of $H$ profiling passes; the cardinality constraint developed in §3.4 renders the choice of solver irrelevant, amortizing the search overhead. *Token-level* allocation (ZipCache (He et al., 2024), KIVI (Liu et al., 2024)) is the finest grain available but introduces runtime overhead from online importance scoring. Importantly, per-head and per-token granularities are orthogonal: a head's bitwidth (static, determined offline) and a token's importance (dynamic, determined online) compose without conflict, so the two can be combined. Per-head therefore occupies the sweet spot for static allocation—fine enough to capture the head-level heterogeneity that drives MCKP benefit on spread-pattern models, yet free of any per-step runtime cost.

Table 6: Effect of compression ratio on Qwen2.5-7B. Constrained MCKP provides meaningful gains at moderate-to-aggressive compression (≥4×). Standard-resolution profiling.

| Compression | Uniform PPL | Constr. MCKP PPL | Δ PPL |
|---|---|---|---|
| 2× | 10.10 | 10.66 | −0.56 |
| 4× | 15.45 | 11.23 | +4.22 |
| 8× | 14,759 | 907 | +13,852 |

## F Budget Sweep

Table 6 shows how the benefit of constrained allocation varies with compression ratio on Qwen2.5-7B. At 2× compression, uniform 8-bit quantization is near-lossless (10.10 PPL vs. 9.81 bf16); constrained MCKP is slightly worse (10.66) due to unnecessary 2-bit assignments. At 4×, constrained MCKP reduces PPL from 15.45 to 11.23, a 27.3% improvement; hires profiling further reduces this to 10.91 (Appendix I). At 8×, both methods degrade catastrophically, but constrained MCKP (907) is an order of magnitude better than uniform (14,759) because the constraint still protects Layer-0 heads.

## G Cross-Dataset Stability

To verify that the profiled allocation is not overfit to the calibration corpus, we profile on C4 and evaluate on WikiText-2, and vice versa, at 4× compression. On Qwen2.5-7B, C4-calibrated MCKP achieves 11.29 PPL on WikiText-2 (vs. 15.45 uniform, a 27% reduction), only 0.06 PPL above the WikiText-2-calibrated result (11.23). In the reverse direction, WikiText-2-calibrated MCKP achieves 20.56 PPL on the C4 held-out split (vs. 35.42 uniform, a 42% reduction; C4 bf16 baseline on this split is 15.32). All three models show >96% bit-assignment agreement across calibration sources: Qwen 96.4% (108/112 heads), Llama 97.7% (250/256), and Mistral 96.5% (247/256). Qwen's agreement is the most informative, as all 112 heads have variable assignments; on Llama, 253 of 256 heads are forced to 4-bit by the $k$=2 constraint, inflating agreement. The cross-dataset results reinforce the heterogeneity-driven narrative: on Qwen, the MCKP benefit is preserved across calibration sources; on Llama and Mistral, the lack of

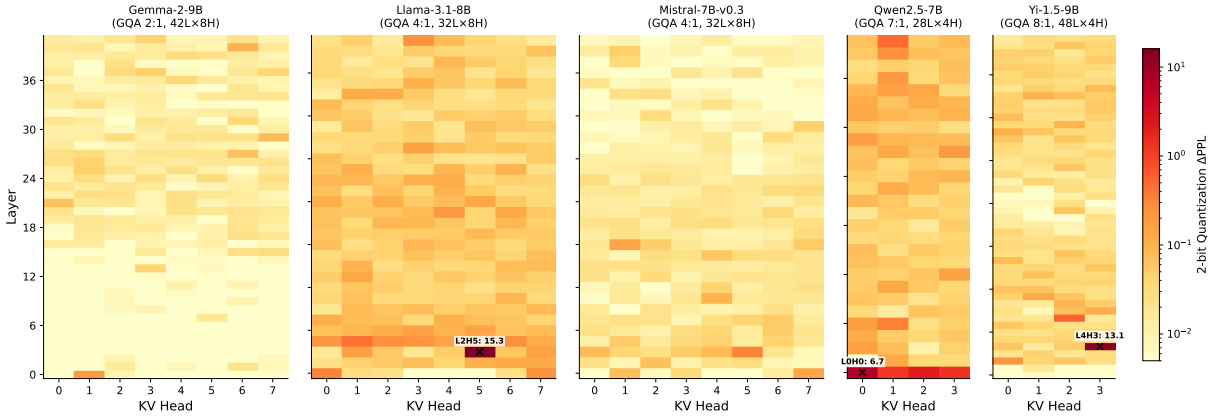



Figure 4: Per-head 2-bit quantization sensitivity across five architectures. Qwen2.5-7B shows high concentration in Layer-0 heads ("spread" pattern); Yi-1.5-9B shows a single extreme outlier at Layer-4 Head-3 ("spike" pattern); Llama-3.1-8B shows a dominant outlier at Layer-2 Head-5 (49% of total distortion, "spike" pattern); Mistral-7B-v0.3 shows near-zero sensitivity across all heads; Gemma-2-9B exhibits the most uniform distribution, consistent with its low GQA ratio (2:1). Color intensity indicates PPL increase when the head is quantized to 2-bit.

benefit is equally stable—Llama's C4-calibrated MCKP is 0.18 PPL worse than uniform, and Mistral shows similar degradation ($-0.03$ to $-0.05$ PPL).

## H Aggressive Compression

At $4\times$ compression, Llama and Mistral show minimal benefit from non-uniform allocation. At more aggressive ratios, the picture changes: at $5.3\times$ (average 3 bits), constrained MCKP achieves $45\times$ lower PPL than uniform on Llama (57.60 vs. 2643.77) and $44\times$ on Mistral (11.91 vs. 519.11), because the constraint protects sensitive heads from 2-bit assignment. At $8\times$, the budget is too constrained for any allocation to preserve quality (both methods degrade catastrophically). The intermediate regime ($5.3\times$) provides the largest relative benefit.

## I Profiling Resolution and Stability

We compare standard-resolution profiling (16 sequences $\times$ 512 tokens) against high-resolution profiling (64 sequences $\times$ 2048 tokens) on Qwen2.5-7B at $4\times$ compression.

High-resolution profiling improves constrained MCKP from 11.23 to 10.91 PPL ($-0.32$ absolute). Relative to the bf16 baseline (9.81), this reduces the degradation from 1.42 to 1.10, a 22.5% reduction. The improvement comes from finer-grained sensitivity estimates: standard profiling detects non-zero 2-bit distortion in 37 of 112 heads, while high-resolution profiling detects 100 of 112 ($2.7\times$ more).

The additional signal changes 8 of 112 head assignments. High-resolution profiling protects shallow layers (L1–L3) at higher bit-widths while compressing mid-layers (L5–L11) more aggressively—a redistribution that standard profiling misses because its distortion estimates for these heads fall below the noise floor.

**Multi-seed stability.** The profiled allocation is stable across calibration subsets. Under hires profiling on Qwen2.5-7B, constrained MCKP yields mean PPL $10.99 \pm 0.06$ across 3 random calibration seeds (individual runs: 11.02, 10.90, 11.05), with std/mean below 0.6%. The bit distribution is identical across all seeds ({2-bit: 8, 4-bit: 100, 8-bit: 4}), and 101 of 112 heads (90.2%) receive the same assignment in all three runs. This stability confirms that the per-head sensitivity ranking is robust to calibration sampling noise.

**Numerical precision.** Float32 profiling ($64 \times 2048$ tokens) yields an identical constrained allocation and PPL within seed variance (10.89 vs. 10.91, $\Delta = 0.02 < \sigma = 0.06$), confirming that bfloat16 precision is sufficient for the profiling step.

**Per-layer optimality.** To validate the additive distortion assumption underlying the MCKP formulation, we compare constrained MCKP against a per-layer brute-force oracle that exhaustively enumerates all $4^4 = 256$ bit-width combinations within each 4-head group on Qwen2.5-7B (hires profiling). Of the 8 layers admitting multiple feasible allocations, MCKP matches the oracle ex-

actly in 5; the remaining 3 show gaps of 0.018 (Layer 1), 0.014 (Layer 11), and 0.005 (Layer 4) PPL. All three gaps involve repositioning a 2-bit assignment among heads with near-identical sensitivity within the same layer bit budget. The negligible total gap ($< 0.04$ PPL) confirms that the per-head additive distortion model closely approximates the true layer-level interaction.

## J KV Separation Ablation

We test whether treating K and V caches as independent items in the MCKP solver—allowing different bit-widths for K and V within the same head—improves over joint allocation where K and V share a bit-width.

Joint allocation achieves 10.91 PPL; separated K/V allocation achieves 11.22 (+0.31 worse). The degradation stems from K-side over-compression: the separated solver doubles the maximum number of 2-bit K items to 16 (vs. 8 in the joint formulation), since K and V are counted independently against the cardinality constraint. Meanwhile, V-side distortion is near-zero for most heads, so the solver assigns all V items to 4-bit, concentrating compression entirely on K.

Of the 112 heads, 21 show K$\neq$V asymmetry in the separated allocation, all with K receiving a lower bit-width than V. This systematic asymmetry suggests that K caches are inherently more compressible than V caches, consistent with the observation that key vectors primarily encode positional information while value vectors carry content.

Joint allocation is sufficient under the current constraint formulation. Separated K/V allocation with per-side cardinality constraints—limiting 2-bit assignments independently for K and V—is a direction for future work.

## K Runtime Benchmark

Table 7 reports end-to-end latency for three KV cache configurations. Uniform 4-bit quantization adds 25.7% decode overhead relative to the bf16 baseline (42.46 vs. 33.78 ms/tok), reflecting per-head quantize/dequantize calls at every attention step. Constrained MCKP adds a further 8.5% over uniform (46.09 vs. 42.46 ms/tok), because mixed-precision allocation dispatches different dequantization kernels for 2-bit, 4-bit, and 8-bit heads within the same layer. The MCKP solver itself runs in $\sim$5 ms and is invoked once at initialization, contributing negligibly to end-to-end latency. Both quantized conditions achieve the same $4\times$ KV cache compression (5.25 MB vs. 21 MB at bf16), so the latency difference between uniform and MCKP reflects only the mixed-precision dispatch overhead, not a memory trade-off.

Table 7: PyTorch-level latency on Qwen2.5-7B (single NVIDIA RTX PRO 6000 (96 GB), prompt_len=256, gen_len=128, mean of 10 runs). All conditions store KV in bf16 with simulated per-head quantize/dequantize calls; peak GPU memory is identical across rows.

| Condition | Prefill (ms) | Decode (ms/tok) | Throughp |
|---|---|---|---|
| Baseline bf16 | 45.86 | 33.78 | 29 |
| Uniform 4-bit | 54.25 | 42.46 | 23 |
| Constrained MCKP | 57.03 | 46.09 | 2 |

**Caveat.** These measurements use PyTorch-level per-head `quantize`/`dequantize` calls and do *not* represent optimized CUDA kernel performance. Production inference engines such as vLLM and TensorRT-LLM employ fused attention kernels that batch dequantization with the matrix multiply, amortizing the per-head dispatch cost that dominates the gap between uniform and MCKP in our simulation. The 8.5% overhead reported here should therefore be treated as an upper bound on the mixed-precision penalty; real-world deployment with fused kernels would likely narrow or eliminate this gap. As noted in the Limitations section, we report no real-kernel latency benchmarks: the numbers above quantify simulation overhead, not production inference speedup.

## L Long-Context Generalization

Table 8: PPL at 2K and 8K context length on WikiText-2. Profiles calibrated at 2K (64 seq $\times$ 2048 tok). Yi-1.5 excluded (max position 4096).

| Model | Method | 2K PPL | 8K PPL |
|---|---|---|---|
| Qwen2.5-7B | BF16 baseline | 9.81 | 6.97 |
| | MCKP ($k$=8) | 10.89 | 8.59 |
| | Uniform 4-bit | 15.45 | 77.75 |
| Llama-3.1-8B | BF16 baseline | 8.09 | 5.60 |
| | MCKP ($k$=2) | 8.50 | 6.05 |
| | Uniform 4-bit | 8.47 | 6.04 |

Table 8 extends the evaluation to 8K context length using 2K-calibrated profiles. On Qwen (spread), uniform 4-bit degrades from $1.57\times$ bf16

at 2K to $11.15\times$ at 8K—quantization errors accumulate catastrophically over longer sequences. Constrained MCKP remains robust ($1.11\times$ at 2K, $1.23\times$ at 8K), recovering 97.7% of the quantization loss at 8K vs. 80.8% at 2K. On Llama (spike), both methods remain indistinguishable at 8K (6.05 vs. 6.04), consistent with the 2K result. The diagnostic (concentration + $t_1/t_5$) thus generalizes across context lengths, and the spread-pattern MCKP advantage amplifies with longer sequences. Profiles calibrated at 2K transfer effectively to 8K without re-profiling, providing positive evidence of calibration generalization.